MATA KULIAH UMUM
POLITEKNIK NEGERI BANDUNG

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PROGRAM STUDI	D4 Teknik Telekomunikasi
MATA PELAJARAN	Matematika Terapan 2
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TANDA TANGAN	

1. 30
2. 30
3. 30
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1a. Sebuah objek bergerak dengan kec; $v(t) = 4 - 2t$ cm/det & pada $t=0$, $s=0$.
Tent persa. posisi benda & lengkapi grafik $v(t)$, $a(t)$, & $s(t)$.

→ $v(t) = 4 - 2t$ $\quad v = \frac{ds}{dt}$ $\quad a = \frac{dv}{dt}$

$s(t) = \int v(t)\,dt = \int 4 - 2t\,dt = 4t - t^2 + C$ ... PU

pada $t=0$

$s(0) = 4(0) - (0)^2 + C = 0$

$C = 0$

$s(t) = 4t - t^2 + 0$

$= (-t^2 + 4t)$ cm. Persamaan khusus (PK)

$a(t) = \frac{dv}{dt} = \frac{d\,4-2t}{dt} = (-2)$ cm/s²

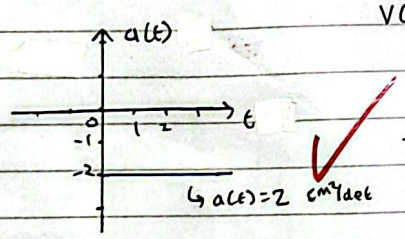
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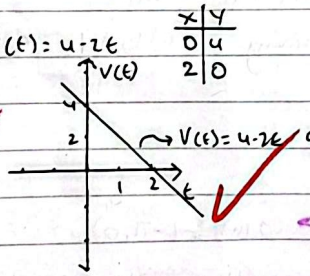
$\rightarrow 2t = 4$

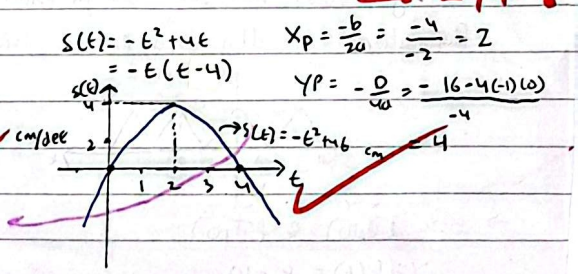
$t = 2$

$\rightarrow -4 + 8$

Gambar grafik







b. $i(t) = 5\cos(2t + \pi/3)$ A, pada $t = \pi/4$, $q(\pi/4) = 0{,}5$. Tent $q(t)$ dari $i(t)$ & $q(t)$ saat $t=0$

→ $i(t) = 5\cos(2t + \pi/3)$ $\quad i = \frac{dq}{dt}$

$q(t) = \int i(t)\,dt = \int 5\cos(2t + \pi/3)\,dt$

$= \frac{5}{2}\sin(2t + \pi/3) + C$ PU

saat $t=0$

$i(0) = 5\cos(2 \cdot 0 + \pi/3) = 5\cos \pi/3$

$= 5 \cdot \frac{1}{2} = 2{,}5$ Ampere

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saat $t = \pi/4$

$q(\pi/4) = \frac{5}{2}\sin\left(2 \cdot \frac{\pi}{4} + \pi/3\right) + C = 0{,}5$

$\frac{5}{2}\sin 5\pi/6 + C = 0{,}5$

$\frac{5}{2} \cdot \frac{1}{2} + C = 0{,}5$

$C = 0{,}5 - \frac{4}{5} = -0{,}3$ X

$q(t) = \frac{5}{2}\sin(2t + \pi/3) - 0{,}3$ Coulomb PK X

saat $t=0$

$q(0) = \frac{5}{2}\sin(2 \cdot 0 + \pi/3) - 0{,}3$

$= \frac{5}{2}\sin \pi/3 - 0{,}3$

$= \frac{5}{2} \cdot \frac{1}{2}\sqrt{3} - 0{,}3$

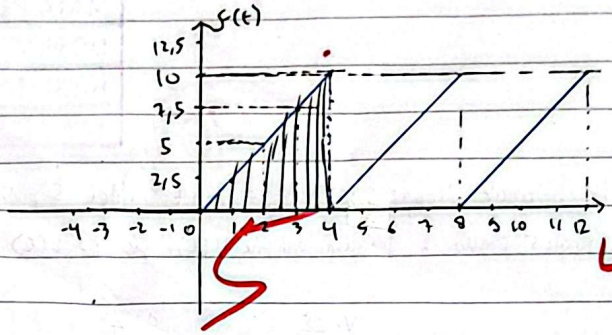
$= \frac{5}{4}\sqrt{3} - 0{,}3$

$= 1{,}865$ C X

SIDU

Dipindai dengan CamScanner

2a $y = f(t) = 2,5t \quad 0 < t < 4$. Sketsa $f(t)$ utk 3 gelombang jika $T = 4$
Hitung Nilai Avg & RMS y



→ t dalam rentang $0 < t < 4$

$t = 0, \quad f(t) = 0$

$t = 1, \quad f(t) = 2,5(1) = 2,5$

$t = 2, \quad f(t) = 5$

$t = 3, \quad f(t) = 7,5$

$t = 4, \quad f(t) = 10$

→ Nilai Average $y = \frac{1}{T}\int_0^T f(t)\,dt$

$= \frac{1}{4}\int_0^4 2,5t\,dt$

$= \frac{1}{4}\,\frac{2,5}{2}t^2\Big|_0^4$

$= \frac{1}{4}\cdot\left(\frac{2,5}{2}\cdot 16\right) = 5$

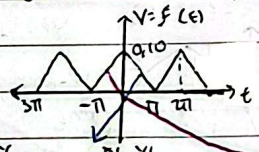
→ Nilai RMS $y = \sqrt{\frac{1}{T}\int_0^T f(t)^2\,dt}$

$= \sqrt{\frac{1}{4}\int_0^4 (2,5t)^2\,dt}$

$= \sqrt{\frac{1}{4}\left[6,25t^2\Big|_0^4\right]}$

$= \sqrt{\frac{1}{4}(6,25\cdot 16)} = 5$

b. Tegangan (V); $V = f(t)$ fungsi periodik dgn $T = 2\pi$
Definisikan $f(t)$ dlm satu periode, hitung nilai RMS V.



→ titik $(0,10)$ & $(\pi,0)$ — titik $(0,10)$ & $(-\pi,0)$

$m = \frac{y-y_1}{x-x_1} = \frac{10-0}{0-\pi} = -10/\pi$

$m = \frac{y-y_1}{x-x_1} = \frac{10-0}{0-(-\pi)} = \frac{10}{\pi}$

titik $(0,10)$: $y = mx + c$

$10 = -10/\pi(0) + c$

$c = 10$

$y = -10/\pi x + 10$

$f(t) = -10/\pi t + 10$

titik $(0,10)$: $y = mx + c$

$10 = \frac{10}{\pi}(0) + c$

$c = 10$

$y = \frac{10}{\pi}x + 10$

$f(t) = \frac{10}{\pi}t + 10$

1 Periode → $f(t)$ $\begin{cases} -10/\pi\, t + 10 & \text{jika } -\pi < t < 0 \\ 10/\pi\, t + 10 & \text{jika } 0 < t < \pi \end{cases}$

→ Nilai RMS $V = \sqrt{\frac{2}{T}\int_0^{T/2} f(t)^2\,dt} = \sqrt{\frac{2}{2\pi}\int_0^{\pi}\left(\frac{10}{\pi}t+10\right)^2 dt}$

$= \sqrt{\frac{1}{\pi}\int_0^{\pi}\left(\frac{100}{\pi^2}t^2 + \frac{200t}{\pi} + 100\right)dt}$

$= \sqrt{\frac{1}{\pi}\left(\frac{100\,t^3}{\pi^2\cdot 3} + \frac{200}{\pi\cdot 2}t^2 + 100t\Big|_0^{\pi}\right)}$

$= \sqrt{\frac{1}{\pi}\left(\frac{100\pi^3}{\pi^2\cdot 3} + \frac{200\pi^2}{\pi\cdot 2} + 100\pi\right)}$

$= \sqrt{\frac{1}{\pi}\left(\frac{100\pi}{3} + 100\pi + 100\pi\right)}$

$= \sqrt{200 + \frac{100}{3}}$

$= \sqrt{\frac{700}{3}}$

$= \frac{10\sqrt{21}}{3}$ V

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3. $0{,}5\frac{di}{dt} + 2i = g(t)$ dgn $i(0) = 0$ Tent $i(t)$ & iss bila a. $g(t) = 10$

b. $g(t) = 10e^{-4t}$

→ $0{,}5\frac{di}{dt} + 2i = g(t)$

$\frac{di}{dt} + 4i = 2g(t)$

$P = 4 \quad Q = 2g(t)$

a. bila $g(t) = 10$

$\frac{di}{dt} + 4i = 2 \cdot 10$

$\frac{di}{dt} + 4i = 20$

→ $P = 4 \quad Q = 20$

→ faktor integrasi $(M) = e^{\int P dt}$

$= e^{\int 4 dt} = e^{4t}$

→ kalikan M ke persamaan

$e^{4t}\frac{di}{dt} + 4e^{4t} i = 20e^{4t}$

gunakan rumus, $ye^{\int P dx} = \int Qe^{\int P dx} dx$

→ $ie^{4t} = \int 20e^{4t} dt$

$ie^{4t} = 5e^{4t} + C$

$i(t) = 5 + Ce^{-4t}$ PU

saat $i(0) = 0$

$i(0) = 5 + Ce^{-4(0)} = 0$

$C = -5$

$i(t) = 5 - 5e^{-4t}$ PK

$iss = 5$

b. bila $g(t) = 10e^{-4t}$

$\frac{di}{dt} + 4i = 2 \cdot 10e^{-4t}$

$\frac{di}{dt} + 4i = 20e^{-4t}$

→ $P = 4 \quad Q = 20e^{-4t}$

→ faktor integrasi $(M) = e^{\int P dt}$

$= e^{\int 4 dt} = e^{4t}$

→ kalikan M ke persamaan

$e^{4t}\frac{di}{dt} + 4e^{4t} i = 20e^{-4t} \cdot e^{4t}$

$e^{4t}\frac{di}{dt} + 4e^{4t} i = 20$

gunakan rumus $ye^{\int P dx} = \int Qe^{\int P dx} dx$

→ $ie^{4t} = \int 20 dt$

$ie^{4t} = 20t + c$

$i = 20e^{-4t} t + ce^{-4t}$ PU

saat $i(0) = 0$

$i(0) = 20e^{-4(0)}(0) + Ce^{-4(0)} = 0$

$C = 0$

$i(t) = 20e^{-4t} t$ PK

$iss = 0$

SiDU

Dipindai dengan CamScanner